

CYBERSECURITY AND INTEROPERABILITY OF AVIATION SAFETY SERVICE ECOSYSTEM

Madhu Niraula, Collins Aerospace, Cedar Rapids, Iowa, USA

Abstract

As the largest high speed mobile network system, the aviation safety service integrated network provides air-to-ground communication to various operators, Air Navigation Service Providers (ANSPs) and airlines across multiple continents. Maintaining such a large network and mobile nodes comes with challenges. Seamless interoperability underpinned by cybersecurity is essential to successful safety service communication. By ensuring that technologies work together across platforms and geographic locations, we better protect the data that flows throughout our aviation systems. And when data is secure, we're able to confidently extend secure communication to reach aircraft, ANSPs, content providers, airline operation centers, and many more no matter where they receive and send messages across multiple endpoints within an aviation ecosystem across the globe. To provide aviation partners with cutting-edge equipment and to ensure that connected devices and systems meet or surpass security standards, the aviation community needs to be committed to ensuring technologies meet needed security specifications including security standards, the Risk Management Framework (RMF), and compliance to the contemporary security protocols and processes. By performing to these specifications, the aviation community can better enable the system-wide availability and cybersecurity of aviation data, thereby facilitating more timely and effective secure communication. This will require an extensive set of external compliance certifications, attestations to objective evidence of compliance to security regulations (examples include ISO 13485, ISO 14971, ISO 27001/18, SOC2) and product cybersecurity testing certification. This also requires combined cybersecurity testing elements of the established standard for software cybersecurity for network-connectable products, as well as security principles from international standards. This paper discusses the critical components of seamless interoperability underpinned by cybersecurity and related issues for secure data exchanges that ensures the continuum of Internet Protocol Suite (IPS) based

safety service communication networks to the aviation safety service ecosystem.

1. Introduction

The deployment of the IPv6 based safety service network in aviation deals with the integration of the internet protocol based information and communications technologies in aircraft, in air-to-ground, ground-to-ground and ground infrastructure. An example of the ICAO, U.S. and European interest is ensuring sustainable aircraft mobility focused on introducing IPS on existing and next generation aircraft [0]. Designing an IPS application is a complex process, in part because it is necessary to have a communication system with a convenient, stable, secure, and economic distribution of data over high speed mobile nodes. Particular attention is paid to moving aviation ACARS and ATN/OSI based air-to-ground communication to IP based communication [1]. The process of integrating IP mobility protocols with aircraft and ground networks can be explained through its three functionalities, namely, aircraft IP address configuration, IP mobility mechanisms, and forwarding IP packets over air-to-ground and core ground-to-ground networks. In addition, there are challenges of integrating the IP mobility with the aviation networks from communication and security perspectives.

2. Architecture

The aviation safety service communication ecosystem is still evolving and currently being updated in ARINC 858 [1] and ICAO DOC 9896 [2]. Airborne and ground applications have evolved in the application areas of Air Traffic Service (ATS) and Airline Operation Control (AOC) thus the basic architecture of both networks is quite similar using a commercial internet protocol based network communication architecture. A typical end to end layer architecture is shown in Fig. 1. The mobile nodes (aircraft) are generally scattered over the region where aircraft need to exchange messages with airline operation centers or ANSPs. Each of these scattered nodes has the capability to generate

data and receive data from various ANSPs as well as the ground end users. Data can be routed to the ANSPs, airline or ground user by a multi-hop infrastructure architecture via terrestrial or satellite links as shown in Fig. 1.

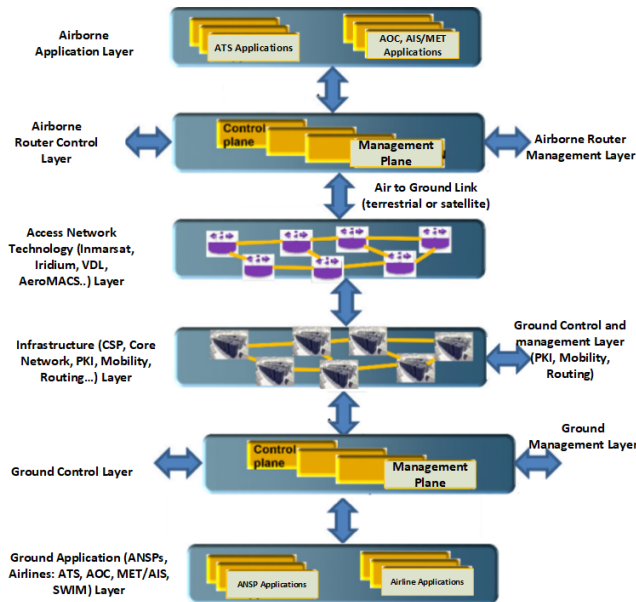


Figure 1: Aviation Safety Service Architecture

Data Plane: This plane includes all data transmission network devices (aircraft, ANSPs and airlines) for applications for the end user for example airborne communication and ground applications between a pilot and controller or airline AOC center application and aircraft flight management system. IPS features a protocol gateway and overlay network to eliminate the heterogeneity of existing safety service networks (ATN/OSI and ACARS). All airborne and ground applications are abstracted as a data plane application. The data plane can be further categorized into a legacy application data plane and a native IP application data plane. Existing ACARS and baseline 1 or baseline 2 [3], [4], [5], [6] applications belong to the legacy data plane, while new applications such as weather service and System Wide Information Management (SWIM) are in the native IP application data plane. Different management policies and adaptations are applied to the airborne router control layer and ground control layer. Airborne application layer and airborne router control layer primarily generate or sync the data plane traffic on the aircraft side. Similarly, the ground application layer and ground control layer generate or sync the data plane traffic on the ground side.

Control Plane: This plane must collect and maintain the status of all the aircraft and is responsible for making packet-forwarding decisions based on aircraft status. The control plane is logically centralized. Control of a network is transferred from an individual access network to the Communication Service Provider (CSP) or core network or regional network. This plane also maintains the current network topology state information, aircraft location, network connectivity, quality of service preferences. Almost all aircraft equipped with IPS capability provide position, aircraft link status and preferences information. The control plane also consists of security provisions, Public Key Infrastructure (PKI) management, routing, and aircraft mobility management.

Management Plane: This plane is responsible for the administration of the other layers.

Application Plane: This plane contains network services and applications for specific content data. Several services compose the application plane such as ATS applications (CPDLC, ADS-C, ATIS, oceanic clearances), weather reports, communication context establishment, security, and Quality of Service (QoS). The services enable or enhance the aviation safety service functionalities.

Communication Interfaces: The control and data planes communicate through a customized interface which includes some predefined control and notification messages. Standard IPS needs to be extended to adapt to aviation safety service communication requirements. Different types of terrestrial and satellite links are abstracted by the airborne router control or ground control layer for the air-to-ground communications. The application plane uses adaptation layer APIs [1], [2] to communicate with the control plane. Unfortunately, there is currently no standardized interface for a consistent communication interface. These are shown as the airborne router management and control layers on the aircraft side with similar ground management and ground control layers on the ground side.

2.1 Network Architecture

An IPS network is an autonomous network. This network must function based on the aircraft movement and based on the link status and QoS preference information from the aircraft without a management center or control center to make its

decisions. The IPS network can decide for itself how it will behave. The aim of IPS is to replace all existing networks (ACARS and ATN/OSI) by a single IP network, thus integrating all communication media. An IPS network must be capable of managing itself, detecting problems, repairing itself and controlling itself.

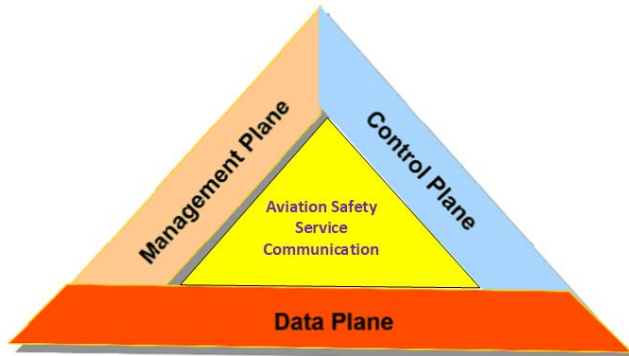


Figure 2: Traditional Network Architecture

For the IPS we need to change the architecture of the network from the traditional sense as depicted in Fig. 2 to a new form of architecture with four planes, where a knowledge plane is added as shown in Fig. 3. The difference with the IPS architecture is the control plane is divided into two sub-planes: a) the control plane and b) the knowledge plane, but the knowledge plane also contains part of the management plane. The management plane has an additional plane in the IPS architecture, but it is integrated into mobility management and PKI infrastructure. Each network (access network, regional network, and core network) has its own knowledge and management plane.

Knowledge Plane: Fig. 3 illustrates the new architecture of the IPS network. The purpose of the knowledge plane is to collect all of the knowledge in the IPS network and for each point in that network to be able to obtain a fairly universal view.

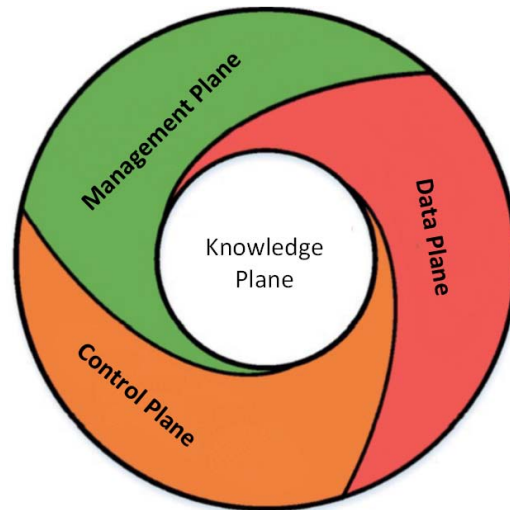


Figure 3: IPS Network Architecture

The knowledge plane can be centralized at a single point or can be distributed to each point. However, a centralized architecture has single point of failure flaws. The purpose of the knowledge plane is to run the control algorithms found in the control plane with aircraft mobility, and to control the data plane [7].

The knowledge plane provides insight to the potential of machine learning and artificial intelligence to address the challenges of future smart aviation networks, where a wide spectrum of approaches, techniques and tools are increasingly receiving attention to tackle issues such as safety, high speed mobility, QoS, strong network dynamic topology and security requirement [8].

2.2 Multihoming / Multilink

Multihoming/multilink involves a node connecting to several networks simultaneously to optimize communications by selecting the best network, or by allowing an application to take several paths. Thus, the packets forming the same message may take several different paths simultaneously. An aircraft connected to VDL mode 2, Inmarsat or Iridium Satellite, AeroMACS, or LDACS networks, allows each application to choose the transport network (or networks) it uses. The term “multihoming” refers to the multiple links that an aircraft may have, with each link connected to a network that can send and receive data traffic simultaneously. Thus, each node has to manage the different access networks and associated links. Fig. 4 illustrates the multihoming/multilink aircraft

connected to multiple access technologies (VDL Mode 2, Satellite and LDACS).

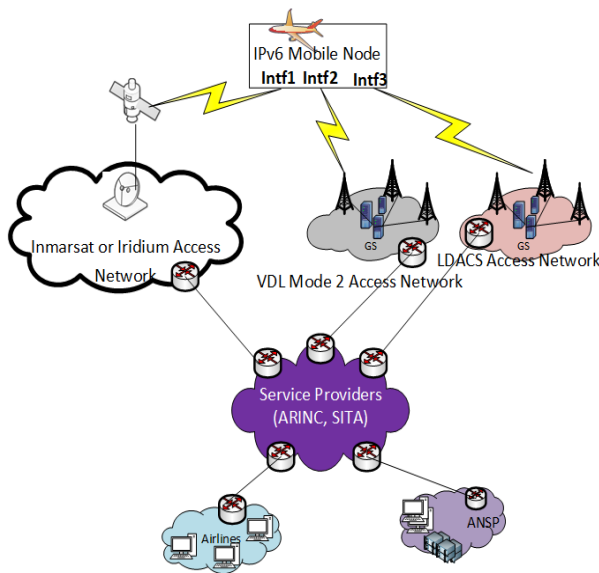


Figure 4: Aircraft as Multihoming/Multilink Mobile Node

Multihoming enables aircraft to have several simultaneous connections to different access networks. Each link can connect to one or more service providers. The goal of this configuration is to improve reliability and optimize performance by using traffic engineering; simultaneously use of multiple air-ground links to increase the bandwidth. The IPS aircraft is the multihoming/multilink node with multiple interfaces (Intf1, Intf2, Intf3) connected to the same or different access network technologies as shown in Fig. 4.

3. Network Topology and Interoperability

An aircraft's points of attachment to the ground might be available in fixed locations while on the ground (airport via AeroMACS) or in the air (e.g., VDL Mode 2, Satellite or LDACS). Thus, offering network access with roaming capability. Figure 5 shows the topology of a Safety Service IPS network and architecture. Fig. 5 shows multiple configurations are possible. The core network is the aviation ground core network and it is connected to multiple regional networks. Since a ground network is a zero trust network, each entry point must manage protections. Some regions may operate their own

access networks, regional network, PKI infrastructure, as well as mobility management.

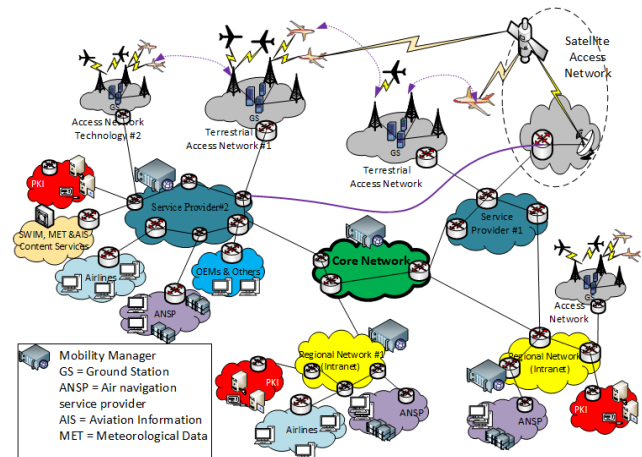


Figure 5: Safety Service IPS End to End Architecture

Since some service providers can over multiple access networks, PKI infrastructure and mobility management, the ANSPs and airlines may use a service provider. Some regions may connect to a regional network, then the regional network can connect to the IPS core network. This complex topology with a zero trust configuration has multiple interoperability and security challenges.

3.1 Interoperability Challenges

There are several open issues and challenges that have not been resolved and require extensive research for the IPS architecture used for aviation. For example, in terms of the IPS environment, the existing air-to-ground mobility signaling and handoff management schemes proposed by ICAO introduce interoperability challenges considering that the aircraft mobility is coordinated and managed by the ground with interoperability among various regions having multiple open issues.

Distributed Mobility Management: As shown in Fig. 5, an aircraft is a high speed mobile node, so it is constantly moving from one access network to the next. In addition, packets sent or received from or to the aircraft are routed to the access network. The ground must track information and update the mobility information in a distributed fashion. A centralized function degrades the network performance because of the traffic connections and possible congestion. In some scenarios, aircraft are moving as a group from one point of attachment to

another; this could peak the traffic and mobility exchanged messages and consequently affect the network performance. Each regional network may choose a different approach, hence impact interoperability of employing a distributed mobility management solution and introduce unforeseen performance problems.

Integration and Interoperability: The system should be able to integrate with existing solutions which are in place already (ANSP ACARS and ATN/OSI End Systems), and airline ARINC 620 host systems at all over the world. Those interfaces are designed to support this existing solution via a protocol gateway [1] and have different complexities. The interoperability with those systems is important, since the IPS should allow data from different sources and systems to be integrated and allow ANSPs and airlines to use existing systems without changes.

Quality of Service: Safety service communication must support various applications with different QoS requirements; for example, safety applications are sensitive to connection and transmission delay. The challenge is how to manage intra/inter domain handoff with a guaranty of the QoS requirements for diverse applications. IPS must combine network resource and application characteristics to the handoff management for achieving efficient resource utilization and seamless roaming. The aircraft and ground must have pertinent knowledge of preference based on the link status information received from the aircraft.

Efficient Handoff Management: The high speed of aircraft especially en-route results in frequent multi-domain handoffs. During the handoff process from a network point of view, many messages are exchanged between network entities and all messages must go through an access network to regional network for registration. On the other hand, from the aircraft point of view, the handoff process suffers from latency and packet loss, which is a burden for most of the applications. Consequently, this will cause degradation in network performance and user QoS.

To meet the safety service application requirements, the handoff management must consider different criteria including handoff latency, throughput, packet loss, signaling cost, control plane overhead and end to end delay. There is no COTS

product that can provide a solution needed for the IPS, nor are COTS vendors willing to adopt any modification in their stack. Handoff management has interoperability issues since it is not clear how the messages are propagated between the access network, regional and core network, as well as end customers.

PKI: For aviation safety service, the ground network is a zero trust network. An aircraft link status and mobility signaling messages as well as application messages require a security provision, since they cross multiple domains. Traditional cyber security mitigation approaches are not adequate and robust enough in offering reliable solutions in aviation networks. Increased connectedness, arrival and departure times of aircraft, and mobility in access networks, make the IPS more vulnerable to cyber-attacks. To realize secure and efficient communication in IPS networks, the following challenges need to be addressed:

Distributed Communication Models: All aircraft are identified and connected through any access network and service provider's servers. Thus, any security breach can throw the entire network into disarray. Each region can have multiple PKI, however, for aircraft to operate seamlessly there is no consistent PKI framework that been identified, but it has been worked at ICAO.

Multilink: The multilink and load-sharing algorithm, which distributes the data along multiple paths, must ensure that the total bandwidth is at least equal to that obtained on the best path. The use of several paths must enhance the reliability of the communication. The multihoming protocol must support mobility by dealing with the problems of horizontal and vertical handovers. Industry specifications still lack for the interfaces to manage multilink for different access technologies that leads to variability in implementation. During validation study, we setup the lab and analyzed the data, analysis shows that many algorithms can create unpredictable scenarios causing bad radio coverage.

3.2 Security

To deploy a secure and reliable IPS network, a set of security requirements must be respected. Failure to adhere to a requirement may lead to an unreliable network and may limit its deployment [7]. It is necessary to ensure the communication and messages exchanged through the network are

generated by legitimate nodes and can't be modified or suppressed by an adversary. The main security requirements related to the IPS are authentication, availability, integrity, non-repudiation, and optional privacy. These requirements can be thought of as criteria to measure the network security. The network should always be available and keep operating for its legitimate nodes even in the presence of faults or attacks. The availability demands high connectivity and continuity to provide fast response times for safety service applications that can't tolerate any delay. The delay makes some messages meaningless or may cause serious consequences [6].

Different types of attacks are highlighted, in addition to how these attacks affect security requirements with a set of countermeasures that could be used against the identified attacks.

3.2.1 Vulnerabilities Related to Mobile Node

The fact that a mobile node's (aircraft) packets can be delivered to any address of visited networks, while it is moving, is an open door, due to possible diversion of traffic to non-legitimate nodes. This type of diversion may be caused by a malicious station that positions itself between the aircraft and one of the other entities communicating with the aircraft. The aircraft can also connect to a false ground network who will then monitor its communications. For example, a false access router may behave as a fake operator and therefore attract victims on this false access network. In other words, during the movement of the aircraft, it is important to secure the mobility and route establishment, the IP address assignment and the autoconfiguration processes. The authentication between all entities of a mobile IP communication model is the first solution to these problems.

3.2.2 Control Plane Security

In the IP networks, the control plane is distributed across millions of lines of code and defined across hundreds of RFCs. The IPS network configuration (see Fig. 5) doesn't have available COTS devices and custom software written for the mobility signaling and management. There is increasing complexity of networks with IP COTS devices making deployment simpler and cheaper than traditional networks (ACARS and ATN/OSI). However, the mobility controller is an entity, which does not exist in traditional networks and its security requires special consideration not available in COTS.

Attackers can remotely differentiate between the IPS and a traditional IP network by analyzing the data traffic and flow response time.

Attacks against the control plane of a IPS network can be categorized into manipulation, availability, and software hacks. In this section the attacks are broken down into more specific threat vectors regarded as the main challenges in securing the IPS network.

3.2.3 Authentication Failure

This authentication attack is analyzed by sending mobility signals from any point on the ground. This attack occurs at the network layer and it aims to compromise the network route information and message data. This type of attack happens when any communicating node is compromised.

Countermeasures: To prevent this attack, authentication is the best possible way, so that no illegitimate node is able to join the network. An authentication service enables a node to prove its identity to any other entity in the network. Without authentication, a malicious node can communicate with other nodes and can easily access unauthorized confidential resources.

3.2.4 Buffer Saturation

The router maintains a memory called Ternary Content Addressable Memory (TCAM). When there is a missing flow rule in the flow table, the router stores part of the packet into buffer memory and sends the header as the packet. When the buffer memory gets full, the router sends the entire packet. The attacker can send numerous packets whose entry does not match in the flow table, therefore, forcing the router to send the entire packet due to the limited capacity of a buffer leading to buffer saturation.

Countermeasures: There are multiple countermeasures documented in literatures [9] and some solutions are implemented by COTS routers such as CISCO routers. These countermeasures are not suitable in IPS because they cannot prevent TCAM exhaustion associated with multipoint mobility signaling flood attacks. However, rate limiting the mobility message will reduce risks.

3.2.5 UDP Flood Attack

UDP flooding attack aims at exhausting the bandwidth resource of the victim machine by diverting numerous UDP packets to it. The attacks which target the bandwidth are not completely curbed

by increasing the bandwidth links of the victim machine; only its resistance can be increased. UDP protocol is a connectionless protocol. In a UDP flood attack, the victim receives numerous UDP packets at different ports. The victim machine then checks for the application on that port; finding none it replies back to the sender with a Destination Unreachable message packet. Due to the absence of any kind of negotiation, spoofing a packet becomes much easier. Using the ICAO DOC 9896 UDP ATNPKT [2] and IP Dialogue Service to analyze this attack, the procedure for executing this attack is straightforward: flooding a network with UDP packets directed to multiple ATS service ports at a high rate. By performing this attack, the victim device must check if any application is listening through that port, if that is not the case, the victim must build an ICMP destination unreachable message and send it to the source of the UDP packet.

Countermeasures: These attacks can be prevented by implementing a traffic policy to limit the rate [10] of received UDP packets, to protect the devices from UDP flooding. If the UDP traffic is limited to a 4000 bytes/s (this value might vary depending on the network infrastructure), the CPU of the device will no longer be overloaded, as it discards any packet that violates the policy.

3.2.6 Security-Default Configurations of Network Devices

Network devices are the foundation of any network infrastructure. They can be managed and configured to fit into different requirements. These devices are delivered with factory settings and that configuration may of be enough to initiate a functional network infrastructure. Default settings are equipped with security holes. All devices must be restricted to prevent unwanted intruders from modifying established settings. Also, device settings should be protected from reading; although writing privileges are thoroughly safeguarded, knowing the exact configuration of the network infrastructure opens multiple possibilities for an attacker being able to research vulnerabilities or security holes on the current settings to subsequently exploit them without the necessity of modifying any network configuration. For these reasons, it is important to count on robust device configurations regarding its access policy and privileges [10]. Otherwise, an

attacker could compromise the established network security policy.

Countermeasures: Determine and establish the correct and right configuration of the network devices. Perform the test and clarify their impact and consequences. Review the established configuration to illustrate their impact on networks and the advantages and disadvantages of their application.

3.2.7 Jamming

With jamming the intruder floods the radio frequencies used by the network with noise and can prevent any exchange of messages. The network can be strongly disrupted if the radio coverage of the intruder is large. The consequence of this attack is a Denial of Service (DoS).

Countermeasures: This area still needs further research. Author observed that government regulated transmitted power may reduce some of the risk. Higher transmitted power implies higher resistance against jamming because a stronger jamming signal is needed to overcome the original signal. A considerable percentage of ground stations currently used in aviation possess the capability to change the output power of their transmitter.

3.2.8 Mobility Vulnerabilities

Mobility signaling is critical for mobility to be operational, but can also be used to set up attacks, if security mechanisms are not provided. Signaling security between the aircraft and the mobility anchor has the possibility of having a fake aircraft mobility anchor as another vulnerability of the mobile IP communication model. A legitimate, but compromised mobility anchor can also be used to inject the wrong routing information in the network, causing all traffic to be redirected, essentially causing multiple security issues.

Countermeasures: Adding a message sequence number plus timestamp, origin authenticity and strong integrity for mobility and aircraft link status signaling messages will eliminate this vulnerability.

3.2.9 Blackhole or Sinkhole

A blackhole or sinkhole is when an adversary is positioned at a routing strategic point of the network and deletes all the messages instead of forwarding them. Thus, the routing service is suspended for all the messages that go through the intruder's node. The adversary captures a message and redirects it to a

remote node of the IPS through a low latency channel. Consequently, a channel is created and messages go through some nodes that should have never seen the messages or that should have seen the messages, but with a greater latency. This attack has a significant influence on routing.

Countermeasures: The prevention to this attack is authentication and integrity protection of all mobility signaling messages between the aircraft and mobility anchor.

3.2.10 Key Management in IPS Networks

IPS is not a spontaneous network as it requires pre-established infrastructure and operational approval of entities in the infrastructure. IPS networks are dynamic in both space and time. They offer large flexibility. However, this flexibility, associated with the vulnerability of wireless communications, makes it necessary to secure data as well as the participating entities. As shown in Fig. 5, there are multiple surfaces an adversary can inject new malicious messages, perform identity usurpation and consequently violate the main security services, namely availability, integrity, authentication and non-repudiation.

Network Security Management: Agbogun, J., and F. A. Ejiga [12] mentioned that different classes of intrusions, network tools, and procedures are employed by attackers in a network hijack. They present how attacks are prevented and minimized at the backend system. The following are the different steps used by intruders during the exploitation of networks. In scanning vulnerable systems, the intruders try to access network connections as well as scan IP addresses of a network. Once hackers can access an IP address, the hackers load Trojan viruses into the network. In addition, intruders use tools to identify vulnerabilities in the network and perform snooping or destroy the operating system. The hacker tries to search the administrative passwords in the computer system and thereby gain access of the entire system.

3.2.11 Authenticity of Mobile Location

The basic security issue in Mobile IP is to prove that the registration messages updating the location of the mobile node are really sent from the mobile node itself (aircraft) and not from a fake node sketching a

man-in-the-middle attack. The authenticity of such messages is important, otherwise it would be extremely easy to implement such an attack or hijacking attack where a machine intruder diverts mobile node traffic by sending a message to update the location instead of the mobile node itself. It is therefore very important to authenticate those entities involved in the operation of registration updates of the mobile node. It is important to authenticate the mobile node and secure its messages to update its location to prevent any fake mobility signaling and causing multiple routing vulnerabilities.

3.2.12 Access Control

Attacks against end systems and any ground node can provide attackers with access to unauthorized resources. This can occur by taking advantage of weaknesses in authentication systems deciding whether communications should be established or not. This can also occur by exploiting system vulnerabilities in communication systems to bypass access control systems of existing resources. Finally, it can also be performed by taking advantage of the lack of proper separation between resources used by different users to monopolize resources and deny access to legitimate users or slow down their operations.

3.2.13 Preventive Mechanisms

The aviation safety networks should be protected against tapping and against the intrusion of some nodes that could spoof the identity of a legitimate node, disrupt routing and availability. Preventive mechanisms make use of cryptographic primitives to guarantee confidentiality, authenticity, integrity, and freshness of information in transit over the network. Each ground segment must be protected.

However, these prevention methods are sometimes ineffective against some attacks, such as the wormhole attack [13]. In addition, there is no assurance that the preventive methods can prevent intrusions. Therefore, other strategies are advocated, such as intruder tolerance and intruder detection. In the first strategy, the network aims to protect itself or reduce the effects of an ongoing attack. In the second strategy, the intrusion is detected and appropriate measures to exclude the intruders are adopted. The second strategy of intruder detection is also interesting because it helps to acquire information on the attack techniques, and thus improves the prevention systems. Intruder detection in the IPS

networks needs to address several scientific challenges. IPS networks are application oriented, i.e. they have very specific characteristics that are dependent on the application they are addressing. The various IPS network configurations and topology make it difficult to model the “normal” or “expected” behavior of the system. Moreover, the methods developed for traditional networks are not applicable to IPS networks, because of the availability of resources being much larger in the IPS networks.

3.2.14 Intruder Detection

For the IPS network given the variability in topology as shown in Fig 5, it is not possible to create a totally infallible defensive mechanism. However, after detecting an intrusion, it is possible to check whether a defensive mechanism has been violated, and then to launch an automatic reaction and to let the network administrator make a decision on an action. In addition, the information provided by an intrusion detection system can be used to improve the defensive mechanisms of the network. In an intrusion detection system, the behavior of the target under protection is controlled and analyzed. Analysis of it assumes that the behavior of the intruders, the normal behavior of the system or the behavior expected from the system are known. According to the class of behaviors under consideration, there are two strategies for detection [14].

Anomaly detection: The observed behavior of the target system is compared to normal and expected behavior. If the behavior of the system is significantly different from the normal or expected behavior, the system is encountering anomalies and is victim of an intrusion.

Misuse detection: The actions undertaken in the target system are compared to the actions usually carried out by intruders and listed in the form of signatures. An intrusion is detected when there is success in identifying a signature from the actions under analysis. The detection of intruders in IPS requires a very different approach from that of conventional IP networks because models, attacks and resources are different. In conventional networks, the role of the user normally exists; the user is the one who uses the network and who generates the traffic profile. In an aviation IPS network, events are monitored by avionics nodes that generate data and send them to a place where ground end systems,

users, airline or ANSPs or ground networks use them. The behavior of the user, in an intruder detection context, is not interesting because the user has no influence on the behavior of the network.

3.2.15 Business Continuity and Disaster Recovery

The aviation safety service core concepts are the same as traditional IP networks: build, test, deploy, and test policies, test procedures, and test technology. These concepts allow the IPS PKI operations to continue to function in the event some type of disaster strikes. The National Institute of Standards and Technology (NIST) has produced a very comprehensive document on business continuity and disaster recovery planning in Special Publication 800-34 [15] Contingency Planning Guide for Federal Information System. However currently what framework will be used for IPS is unclear, it may be the case that each region may choose a different approach.

3.2.16 Incident Management

A PKI compromise might occur and the steps necessary to prepare for such an incident is needed. Incident management includes preparing for a PKI breach incident and having a response plan to execute in the event of a breach.

Offline ROOT CA: In August 2012, the NIST released an update to its Computer Security Incident Handling Guide, Special Publication 800-61 [16]. This current guide addresses new technologies and attack vectors and changes the prioritization criteria for incident response and facilitates information sharing. There is currently no specific section on PKI incidents for the IPS. ICAO is also missing the provision related to this issue.

3.2.17 PKI and Certificate Authority (CA)

When a PKI system issues a certificate to an unconfirmed business identity, the CA must assume liability. Currently provisions must be made to support continued operation when PKI and CA systems fail, or a security breach occurs. For example, when the now bankrupted Dutch CA DigiNotar was compromised in 2011, fraudulent certificates were issued for Google, Yahoo, Mozilla, and others [17] prompting unwanted implications of warranties, liabilities, or indemnifications. The avionics device ownership might be different for an ANSP, airline and other organizations. While the

RFC 3647 section 3.2 addresses requirements and rules for protecting system and network components including hardware, software, applications, and information objects various aspects of key management are not discussed. It is still unclear what common framework can be used for aviation where CA and PKI failure may have some safety impacts.

3.2.18 PKI Governance Risk and Compliance

For any organization to be successful with governing its PKI systems, there are three essential support groups whose cooperation and interactions are crucial: management, security, and audit. For the IPS currently all these parts are still evolving, and no final framework has been formed yet. This area requires more research and analysis in the coming year based on the ongoing work in ICAO, RTCA, EUROCAE and AEEC standards development organizations.

3.2.19 Intrusion Tolerance

In the context of the IPS, routing has most of the work specified on intrusion tolerance. For intrusion tolerance the idea is to make critical functions of the system as resistant as possible to any compromising attacks by an intruder. Several works define multiple routes for simultaneous or alternative usages, to guarantee full or partial delivery of messages [19]. Some other works attempt to establish new routes once communication problems are detected [20].

3.2.20 Audit and Compliance

In the IPS the control of storage overhead is mandatory within the networks and avionics. However currently there is no common framework exists that can be concisely applied to IPS for the aviation safety service ecosystem.

3.2.21 Supply Chain Risk

Managing cyber risks in supply chains presents challenges for the aviation safety service communication, especially in times where every organization tries to combat cyber risks for supply chains. There is the need to enlarge the view from a single company's point of view to the supply-chain view for the IPS. Author has analyzed supply chain cyber risks for the IPS based safety service communication in order to shed light into the underlying topic. This will be explored further in future research.

3.2.22 Legal Analysis—Notification Duty

As shown in Fig. 5, multiple entities are connected to the service provider. The relevant question is what the service provider must do or rather is allowed to do when the knowledge about a vulnerability comes from the domain of another customer. In this case, the service provider finds itself in the peculiar situation to potentially breach one of the contracts, regardless of whether it decides to act or to refrain therefrom. For example, while providing services to customer A, the service provider becomes aware of IP addresses of botnet members or DDoS servers within the domain of ANSP customer XYZ. Is it lawful to share this information with ANSP customer ABC to mitigate risks for ANSP XYZ? When a service provider has knowledge of a severe vulnerability, should it be obliged to mitigate the risk for all its customers. Otherwise, the service provider would not be taking actions to protect one of its customers despite better knowledge, which would most likely lead to the service provider's liability due to breach of contract.

However, the scope and the details of the service provider's duties are the object of negotiation and may differ by what is agreed on in a contract and particularly in the Service Level Agreement (SLA). A service provider may have customers connected from multiple countries which have their own laws and regulations. Currently there is no common framework to handle and mitigate the risk for the aviation safety service IPS networks.

Conclusion

The IPS network is a promising end to end safety service communication means for a better aviation future. It has been found to provide several prospective objectives which could help in improving air traffic efficiency and safety, as well as providing command and control messaging. IPS networks are still in need of more investigations and countermeasures against different challenges and security attacks that constrain their deployment. In this paper, the IPS network and its structure have been assessed with an overview of the main IPS challenges and security requirements. Different types of attacks have been investigated and highlighted, in addition to how these attacks affect security requirements. Moreover, a set of countermeasures have been studied and surveyed that could be used against the identified attacks. The IPS is

transitioning toward increasing inter-connection between the various networks and systems, which will necessitate a coordinated effort to ensure seamless interoperability while addressing cybersecurity for the aviation safety service communication networks.

References

- [0] DOC 9752, 2019, Global Air Navigation Plan (GANP) - Sixth Edition, Montreal, Canada, International Civil Aviation Organization, https://www.icao.int/publications/Documents/9750_6_ed_en.pdf
- [1] ARINC 858 Part 1 and Part 2, 2021, INTERNET PROTOCOL SUITE (IPS) FOR AERONAUTICAL SAFETY SERVICES PART 1, Maryland, USA, SAE ITC
- [2] International Civil Aviation Organization 2015, Manual on the Aeronautical Telecommunication Network (ATN) using Internet Protocol Suite (IPS) Standards and Protocol, 2nd edition, Montreal, Canada, ICAO
- [3] ARINC 622, 2017, ARINC 622 ATS Data Link Applications over ACARS Air-Ground Network, Maryland, USA, ARINC
- [4] ARINC 623, 2005, ARINC 623, CHARACTER-ORIENTED AIR TRAFFIC SERVICE (ATS) APPLICATIONS, Maryland, USA, ARINC
- [5] ED-110B, 2007, Interoperability Requirements Standard for Aeronautical Telecommunication Network Baseline 1 (Interop ATN B1), Paris, France, EUROCONTROL
- [6] RTCA, 2017, RTCA DO-350 VOLUME 1 & 2 - Safety and Performance Requirements Standard for Baseline 2 ATS Data Communications (Baseline 2 SPR Standard, Washington, DC, RTCA
- [7] Niraula, M., 2019, Security Consideration For The IPV6 Based Air To Ground Safety Service Communication, Integrated Communications, Navigation and Surveillance Conference (ICNS), 2019, pp. 1-11, doi: 10.1109/ICNSURV.2019.8735356.
- [8] Suma, V., Nouredine Bouhmala, and Haoxiang Wang, eds. 2021, Evolutionary Computing and Mobile Sustainable Networks: Proceedings of ICECMSN 2020. Berlin/Heidelberg, Germany: Springer Pages 113-125
- [9] Jeong, H. J., I. S. Song, T.G. Kwon, and Y.K. Lee, 2006, A multi-dimension rule update in a team-based high-performance network security system. In 20th International Conference on Advanced Information Networking and Applications-Volume 1 (AINA'06) (Vol. 2, pp. 62-66). IEEE.
- [10] Singh, A. D. Juneja, 2010, Agent based preventive measure for UDP flood attack in DDoS attacks. International Journal of Engineering Science and Technology, 2(8), 3405-3411.
- [11] Strassner, J, J. S. Strassner, J. S. 2004, Policy-based network management: solutions for the next generation. Chapter 3 & 4, Morgan Kaufmann.
- [12] Agbogun, J., and F. A. Ejiga. 2013, Network Security Management. Netw. Secur 2.4, pages 617-625.
- [13] Fuentes-García, M., J. Camacho, and G. Maciá-Fernández, 2021. Present and Future of Network Security Monitoring. IEEE Access, 9, pp.112744-112760.
- [14] Smys, S., A. Basar, and H. Wang, 2020, Hybrid intrusion detection system for internet of things (IoT). Journal of ISMAC, 2(04), 190-199.
- [15] Swanson, Marianne, Pauline Bowen, Amy Wohl Phillips, Dean Gallup, David Lynes, 2010, Contingency Planning Guide for Federal Information Systems, NIST Special Publication 800-34 Rev. 1, National Institute of Standards and Technology, <https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-34r1.pdf>
- [16] Cichonski, Paul, Tom Millar, Tim Grance, 2012,

Karen Scarfone, 2012, Computer Security Incident Handling Guide Recommendations of the National Institute of Standards and Technology, NIST, <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-61r2.pdf>

[17] Wolff, Josephine, and Sandra Braman. 2018, 5 Certificates Gone Rogue: The DigiNotar Compromise and the Internet's Fragile Trust Infrastructure, page 81-100.

[18] Chokhani, S., Ford, W., Sabett, R., Merrill, C. & Wu, S., 2003, Internet X.509 Public Key Infrastructure Certificate Policy and Certification Practices Framework”, RFC 3647. Available at: <https://www.ietf.org/rfc/rfc3647.txt>, IETF

[19] Babay, A., J. Schultz, T. Tantillo, S. Beckley, E. Jordan, K. Ruddell and Y. Amir, 2019, Deploying intrusion-tolerant SCADA for the power grid. In 2019 49th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN) (pp. 328-335). IEEE

[20] Nuo, Y., 2018, A novel selection method of network intrusion optimal route detection based on naive Bayesian. International Journal of Applied Decision Sciences, 11(1), 1-17.

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Email Addresses

Madhu.Niraula@collins.com

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